

Vivaswat Shastry

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Summary

- Developed 3 software packages (in C++, R & python) of novel statistical & machine learning methods for inferring biological processes with >20,000 downloads in total
- Led statistical modeling & complex data analysis in 5 peer-reviewed publications ([Google Scholar](#))
- Engineered machine learning pipelines for prediction & inference from messy data across various fields for 6+ years
- Designed & delivered workshops in scientific computing & data visualization with R, python, Unix shell & git for 8+ years as a Certified Software & Data Carpentry Instructor (from [The Carpentries](#))
 - Optimized workflows daily on high-performance computing (HPC) cluster systems
- Amateur stand-up comedian

Projects

University of Chicago, Chicago, IL

Sep 2020 - present

Department of Human Genetics, Graduate Research Assistant

- Conducted a massive simulation study (**100,000 replicates**) to assess the impact of extra variables in parameter inference (published in [GENETICS](#))
 - Computed Fisher Information & KL divergence to measure variable interpretability
 - Developed a massively parallel data processing pipeline to run in **<30 minutes**
- Implemented novel method to model outliers in a graph-based framework to derive insights from spatial data (package with user-friendly documentation: [FEEMSmix](#))
 - Decreased computational complexity of algorithm to obtain **>4x speedup** with **5% increase in R²** over current state-of-the-art method

University of Wyoming, Laramie, WY

Jan 2018 – Aug 2020

Department of Botany, Graduate Research Assistant

- Developed novel statistical model for ancestry & genotype estimation in low-coverage sequencing data to mixed-ploidy samples using C++ (package with vignette: [popgen-entropy](#))
- Improved runtime (**10x speedup**) and accuracy (**30% increase in precision**) of a hierarchical Bayesian model with STAN compared to JAGS (CRAN package: [CNVRG](#))

Education

University of Chicago, Chicago, IL

Ph.D. in Computational Biology (with a focus in Genomics)

Sep 2025 (expected)

University of Wyoming, Laramie, WY

M.S. in Botany

Aug 2020

B.S. in Electrical Engineering, Minor in Computational Science

Dec 2017

Honors: *summa cum laude*, Tau Beta Pi

Technical Skills

- **Statistics:** multiple linear regression, linear mixed models, empirical Bayes
- **Machine Learning:** decision trees/random forest, variational autoencoders, unsupervised clustering techniques (PCA, k-means, LDA), cross-validation
- **Programming:**
 - **python** (NumPy, SciPy, sklearn, statsmodels, pandas, plotly, seaborn, matplotlib, PyTorch)
 - **R** (dplyr, ggplot2, tidyr, sp, STAN, Rmarkdown, Shiny)
 - **C++** (GSL, OpenMP, MPI)
- **Version Control & Workflow:** git/GitHub, *nix, Bash, SLURM, Jupyter, snakemake, Google Suite